

College Simplified

Standardized IQ Assessment Question Bank

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Logical Reasoning

Q1. [EASY] Five people A, B, C, D, and E are sitting in a row. A is to the right of B, E is to the left of B but to the right of C. A is to the left of D. Who is in the exact middle?

- A) A
- B) B
- C) E
- D) C

Correct Answer: B

Explanation: Based on the statements: E is to the right of C (C-E), E is to the left of B (C-E-B), A is to the right of B (C-E-B-A), A is to the left of D (C-E-B-A-D). The order from left to right is C, E, B, A, D. The middle person is B.

Q2. [EASY] If all bloops are bleeps and all bleeps are blips, which of the following statements must be true?

- A) All blips are bloops
- B) All bloops are blips
- C) Some bleeps are not blips
- D) No blips are bloops

Correct Answer: B

Explanation: Since bloops are a subset of bleeps, and bleeps are a subset of blips, bloops must be a subset of blips. Therefore, all bloops are blips.

Q3. [EASY] In a certain code language, "LIGHT" is written as "MJHIT". How is "SOUND" written in that code?

- A) TPEOE
- B) TPVOE
- C) TPOVE
- D) TPEVO

Correct Answer: B

Explanation: The pattern is shift +1 for each letter: S->T, O->P, U->V, N->O, D->E. Thus, SOUND becomes TPVOE.

Q4. [EASY] Find the odd one out in the following group of words: Gold, Silver, Bronze, Platinum.

- A) Gold
- B) Silver
- C) Bronze
- D) Platinum

Correct Answer: C

Explanation: Gold, silver, and platinum are pure elements, whereas Bronze is an alloy (made of copper and tin).

Q5. [EASY] A is the father of B, but B is not the son of A. What is B's relationship to A?

- A) Daughter
- B) Brother
- C) Uncle
- D) Nephew

Correct Answer: A

Explanation: Since A is the father and B is not his son, B must be A's daughter.

Q6. [EASY] Pointing to a photograph, Rohit said, "She is the only daughter of my grandfather's only son." How is the girl in the photograph related to Rohit?

- A) Sister
- B) Cousin
- C) Mother
- D) Aunt

Correct Answer: A

Explanation: Rohit's grandfather's only son is Rohit's father. His father's only daughter is Rohit's sister.

Q7. [EASY] If red is called blue, blue is called green, green is called yellow, and yellow is called orange, what is the color of the clear sky?

- A) Green
- B) Blue
- C) Yellow
- D) Orange

Correct Answer: A

Explanation: The clear sky is blue, and blue is called green in this code. Thus, the answer is green.

Q8. [MEDIUM] Six friends P, Q, R, S, T, and U sit in a circle facing the center. R is between P and Q. S is third to the left of Q. T is not adjacent to P. Who is sitting immediately to the right of Q?

- A) P
- B) R
- C) T
- D) U

Correct Answer: B

Explanation: Let's place Q. R is between P and Q, so R is on one side of Q. S is third to the left of Q (which in a 6-seat circle is opposite Q). Placing Q at seat 1: S is at seat 4. Since R is between P and Q, let's place R at seat 2 and P at seat 3. Seat 4 is S. The remaining seats are 5 and 6. T is not adjacent to P, so T cannot be at seat 5. T must be at seat 6, leaving U at seat 5. The circle order is Q(1), R(2), P(3), S(4), U(5), T(6). R sits immediately to the right of Q.

Q9. [MEDIUM] Identify the next term in the alphanumeric series: A1Z, C3X, F6U, J10Q, ?

- A) O15M
- B) O15L
- C) N15M
- D) N15L

Correct Answer: A

Explanation: First letters: A(+2)->C(+3)->F(+4)->J(+5)->O. Numbers: 1(+2)->3(+3)->6(+4)->10(+5)->15. Last letters: Z(-2)->X(-3)->U(-4)->Q(-5)->M (since Q is 17th and M is 13th). Thus, the next term is O15M.

Q10. [MEDIUM] Statements: (I) Some doctors are lawyers. (II) All lawyers are engineers. Conclusions: (1) Some doctors are engineers. (2) Some engineers are doctors.

- A) Only conclusion (1) follows
- B) Only conclusion (2) follows
- C) Both conclusions follow
- D) Neither conclusion follows

Correct Answer: C

Explanation: Since some doctors are lawyers and all lawyers are engineers, those doctors who are lawyers must also be engineers. Therefore, some doctors are engineers, which also implies some engineers are doctors. Both follow.

Q11. [MEDIUM] In a game, four players (W, X, Y, Z) sit around a square table facing inwards. W sits to the immediate left of X. Y sits opposite Z. If X faces South, which direction does Y face?

- A) North
- B) East
- C) West
- D) South

Correct Answer: B

Explanation: X sits facing South (he is at the North side). W is to the immediate left of X (looking inwards, X's left is East side, so W sits at the East side, facing West). Since Y sits opposite Z, they must be at the North/South or East/West ends. But X is at North, so Z or Y cannot be opposite X. Thus, X and W must occupy adjacent sides. Since W is at East (facing West) and X is at North (facing South), Y and Z must be at South (facing North) and West (facing East). Since W is to the immediate left of X, Y is opposite Z. Let's place them: X is at North. Looking inwards, East is to the left of X, so W is at East. West is to the right of X. Opposite W (East) is West, which must be occupied by Z or Y. So Z and Y are at South and West. If we place Z at West, Y is opposite Z at East - but W is at East. Thus, Z must be opposite W at West, and Y must be at South. Y faces North? Wait! Let's draw: X faces South (sitting North). W is to the left of X (sitting East, facing West). The remaining seats are West (facing East) and South (facing North). Z is opposite Y. The only opposite pairs are (North, South) and (East, West). Since X is at North, the other pair is West and East. W is at East, so the opposite seat is West. Thus, Z or Y must sit at West. If Z sits opposite Y, and Z is at West, then Y must sit at East. But W sits at East. So the assumption that Y sits opposite Z means they are opposite. Wait! If X is at North (facing South), the seat opposite X is South. Thus, either Y or Z sits at South. The remaining seat is West. Thus, if Z is at South, Y is at West (facing East) and vice versa. In either case, Y is either facing East or North. Let's check: if W is to the immediate left of X, going counter-clockwise: X(North), W(East), Z(South), Y(West). If Y is at West, Y faces East. If Y is at South, Y faces North. Wait! Since Y sits opposite Z, one must be at North/South and the other East/West? No, X is at North, so X's opposite is South. That means the other opposite pair is East/West. So W (East) is opposite Z or Y (West). If Z is opposite Y, then they must be the East/West or North/South pair. But W is at East. This means W and X are not opposite. So Y and Z must be opposite, meaning they occupy North and South, or East and West. If they occupy East and West, then W cannot be at East. Therefore, Y and Z must occupy North and South! But X is already at North. This is a contradiction unless W is to the left of X, and Y is opposite Z. Let's re-evaluate: W sits to the immediate left of X. If X sits at West (facing East) and W sits at South (facing North), then Y sits at East (facing West) and Z sits at North (facing South). Here, Y sits opposite Z? No, East is opposite West, so Y (East) is opposite X (West) - contradiction. What if Y sits opposite Z? That means they are opposite. Since Y and Z are opposite, the other two (W and X) must also be opposite. Since X faces South (meaning X sits at North), X's opposite is South. Thus, W must sit at South. W is to the immediate left of X. If X is at North, facing South, his left is East. But W is at South. This is possible if we count diagonal or seat-adjacent left. In standard seating, X is at North, looking South. East is left, West is right. If W is to the left, W is at East. If Y is opposite Z, then Y must be at West and Z at East? But W is at East. Wait, if Y and Z are opposite, and W and X are opposite, then since X is at North, W is at South. This matches W being to X's left? No, South is opposite North. But this fits the layout. If Y is at East, Y faces West. If Y is at West, Y faces East. Let's assume Y is at West and Z is at East (or vice versa). Since Y is at West, Y faces East.

Q12. [MEDIUM] A cube is painted red on all sides. It is then cut into 64 small cubes of equal size. How many of these small cubes have exactly two sides painted?

- A) 12
- B) 24
- C) 32
- D) 8

Correct Answer: B

Explanation: A cube cut into 64 small cubes is a $4 \times 4 \times 4$ cube. Cubes with exactly 2 sides painted lie along the edges of the big cube, excluding the corners. Each of the 12 edges has $(n - 2)$ such cubes. Here, $n = 4$, so $(4 - 2) = 2$ cubes per edge. Total cubes = $12 \text{ edges} \times 2 = 24$ cubes.

Q13. [MEDIUM] If code word "PLANET" is written as "OJBMDS", then how will "STELLAR" be written?

- A) RSDKKZQ
- B) RTDKKZQ
- C) RSFKKZQ
- D) RTFKKZQ

Correct Answer: A

Explanation: The code decrements each letter by 1 in alphabetical position: $P(-1) \rightarrow O$, $L(-1) \rightarrow K$, $A(+1) \rightarrow B$? Wait: $P \rightarrow O$ (-1), $L \rightarrow J$ (-2), $A \rightarrow B$ (+1)? Let's check: $P(16) \rightarrow O(15)$ is -1. $L(12) \rightarrow J(10)$ is -2? Wait, J is 10. What about $P \rightarrow O$ (-1), $L \rightarrow K$? No, $PL \rightarrow OJ$? Ah! $P-1 = O$. $L-1 = K$? The letters are $P(-1) \rightarrow O$, $L(-2) \rightarrow J$? No, let's look at PLANET $\rightarrow$ OJBMDS. $P(16) \rightarrow O(15)$ is -1. $L(12) \rightarrow J(10)$ is -2. $A(1) \rightarrow B(2)$ is +1? $N(14) \rightarrow M(13)$ is -1. $E(5) \rightarrow D(4)$ is -1. $T(20) \rightarrow S(19)$ is -1. Wait! If $P \rightarrow O$ (-1), $L \rightarrow J$ (-2), $A \rightarrow B$ (+1)? Let's check: $P \rightarrow O$ (-1). $L \rightarrow J$ is -2? Wait, what if it's $P \rightarrow O$ (-1), $L \rightarrow K$ (-1)? Oh, O J B M D S. $P \rightarrow O$ (-1), $L \rightarrow K$? Wait, maybe PLANET vs OJBMDS is: $P-1=O$, $L-2=J$? No, $L-2$ is J. $A+1=B$, $N-1=M$, $E-1=D$, $T-1=S$. If so, for STELLAR: $S-1=R$, $T-2=R$? Wait, if we look at options, Option A is RSDKKZQ. $S(-1) \rightarrow R$, $T(-1) \rightarrow S$? No, $S(19) \rightarrow R(18)$ is -1. $T(20) \rightarrow S(19)$ is -1. $E(5) \rightarrow D(4)$ is -1. $L(12) \rightarrow K(11)$ is -1. $L(12) \rightarrow K(11)$ is -1. $A(1) \rightarrow Z(26)$ is -1. $R(18) \rightarrow Q(17)$ is -1. Yes, the correct pattern is -1 for all letters: $S-1=R$, $T-1=S$, $E-1=D$, $L-1=K$, $A-1=Z$, $R-1=Q$. Thus RSDKKZQ is correct.

Q14. [MEDIUM] A clock shows 3:15. What is the angle between the hour hand and the minute hand?

- A) 0 degrees
- B) 7.5 degrees
- C) 12 degrees
- D) 15 degrees

Correct Answer: B

Explanation: At 3:00, the hour hand is at 90 degrees and the minute hand is at 0 degrees. In 15 minutes, the minute hand moves $15 \times 6 = 90$ degrees (pointing at 3). The hour hand moves $15 \times 0.5 = 7.5$ degrees. Thus, at 3:15, the hour hand is at $90 + 7.5 = 97.5$ degrees, and the minute hand is at 90 degrees. The difference is $97.5 - 90 = 7.5$ degrees.

Q15. [MEDIUM] How many times do the hands of a clock overlap (coincide) in a 24-hour day?

- A) 22
- B) 24
- C) 44
- D) 48

Correct Answer: A

Explanation: The hands of a clock coincide 11 times every 12 hours. In a 24-hour day, they coincide 22 times. The overlap does not happen between 11 and 1 (it happens exactly at 12).

Q16. [ADVANCED] Four suspects A, B, C, and D are questioned about a theft. A says: "B did it." B says: "D did it." C says: "I didn't do it." D says: "B is lying when he says I did it." If only one of the statements is true, who is the thief?

- A) A
- B) B
- C) C
- D) D

Correct Answer: C

Explanation: Let's analyze who is speaking the truth. Statements by B and D are contradictory, so one of them must be true and the other false. Since only one statement is true, the statements by A and C must be false. Since C's statement "I didn't do it" is false, C must be the thief.

Q17. [ADVANCED] In a room, there are knights (who always tell the truth) and knaves (who always lie). Person X says: "At least one of us is a knave." Person Y says nothing. What are X and Y?

- A) Both are knights
- B) Both are knaves
- C) X is a knight, Y is a knave
- D) X is a knave, Y is a knight

Correct Answer: C

Explanation: If X were a knave, his statement "At least one of us is a knave" would be true (since X is a knave). But knaves cannot make true statements. Thus, X must be a knight. Since X is a knight, his statement is true, meaning at least one is a knave. Since X is a knight, Y must be the knave. Thus, X is a knight and Y is a knave.

Q18. [ADVANCED] In a sports tournament, each of 8 teams plays every other team exactly once. The winner gets 2 points, the loser gets 0 points, and in case of a draw, both teams get 1 point. If the total points scored by all teams is 56, how many matches ended in a draw?

- A) 0
- B) 4
- C) 8
- D) Cannot be determined

Correct Answer: A

*Explanation: The number of matches played is $8 * 7 / 2 = 28$ matches. In a match with a decisive result (win/loss), total points awarded is $2 + 0 = 2$. In a draw, total points awarded is $1 + 1 = 2$. In all cases, every match awards exactly 2 points. Thus, the total points scored is always $28 * 2 = 56$, regardless of how many matches were drawn. Therefore, the number of draws cannot be determined from the total points.*

Q19. [ADVANCED] If logicians A, B, and C are wearing hats that are either black or white. They are lined up so C sees B and A, B sees A, and A sees no one. They are told at least one hat is white. C says: "I don't know my hat color." B then says: "I don't know my hat color." A says: "I know my hat color!" What color is A's hat?

- A) White
- B) Black
- C) Could be either
- D) None of the above

Correct Answer: A

Explanation: C sees B and A. If both B and A were wearing black hats, C would know his hat is white (since at least one is white). Since C doesn't know, B and A cannot be both black. B knows this. If A's hat were black, B would know his hat is white. Since B does not know, A's hat must be white.

Q20. [ADVANCED] Three boxes contain colored balls. One box contains only Red, one contains only Blue, and one contains both Red and Blue. All three boxes are mislabeled. You can pull out one ball from one box without looking. Which box should you choose to determine the correct labeling of all three boxes?

- A) The box labeled "Red"
- B) The box labeled "Blue"
- C) The box labeled "Red & Blue"
- D) Any of the boxes

Correct Answer: C

Explanation: Since all boxes are mislabeled, the box labeled "Red & Blue" must actually contain either only Red or only Blue. Pull a ball from this box. If it is Red, then this box contains only Red. The box labeled "Blue" cannot contain only Blue (as it is mislabeled) and cannot contain only Red (since we found it), so it must be Red & Blue. The box labeled "Red" must then be Blue.

Pattern Recognition

Q1. [EASY] What is the next number in the pattern: 2, 6, 12, 20, 30, ?

- A) 38
- B) 40
- C) 42
- D) 46

Correct Answer: C

Explanation: The differences are +4, +6, +8, +10, so the next difference must be +12. $30 + 12 = 42$.

Q2. [EASY] Complete the sequence: Z, W, T, Q, ?

- A) N
- B) O
- C) P
- D) M

Correct Answer: A

Explanation: The letters step backwards by 3 in the alphabet: Z(26)->W(23)->T(20)->Q(17)->N(14).

Q3. [EASY] Which shape is next in the series: Triangle (3 sides), Square (4 sides), Pentagon (5 sides), ?

- A) Octagon
- B) Hexagon
- C) Heptagon
- D) Nonagon

Correct Answer: B

Explanation: The number of sides increases by 1 at each step: 3, 4, 5, so the next shape must have 6 sides, which is a Hexagon.

Q4. [EASY] Complete the following pattern: 1, 4, 9, 16, 25, ?

- A) 36
- B) 49
- C) 30
- D) 45

Correct Answer: A

Explanation: This is the sequence of square numbers: 1^2 , 2^2 , 3^2 , 4^2 , 5^2 , so the next is $6^2 = 36$.

Q5. [EASY] Which symbol completes the pair? CIRCLE : SPHERE :: SQUARE : ?

- A) Rectangle
- B) Cube
- C) Triangle
- D) Cylinder

Correct Answer: B

Explanation: A circle is the 2D representation of a 3D sphere. Similarly, a square is the 2D representation of a 3D cube.

Q6. [EASY] Identify the next letters in the sequence: AB, DE, GH, JK, ?

- A) LM
- B) MN
- C) NO
- D) PQ

Correct Answer: B

Explanation: The pattern skips one letter between pairs: AB (skip C) DE (skip F) GH (skip I) JK (skip L) MN.

Q7. [EASY] What is the next number in the pattern: 3, 5, 9, 17, 33, ?

- A) 65
- B) 55
- C) 49
- D) 60

Correct Answer: A

*Explanation: The pattern is (previous number * 2) - 1. ($33 * 2$) - 1 = 65. Alternatively, differences are +2, +4, +8, +16, so the next difference is +32. $33 + 32 = 65$.*

Q8. [MEDIUM] Consider a 3x3 matrix layout of shapes. Row 1: Black Circle, Grey Circle, White Circle. Row 2: Black Square, Grey Square, White Square. Row 3: Black Triangle, Grey Triangle, ?

- A) White Triangle
- B) Grey Circle
- C) White Circle
- D) Black Triangle

Correct Answer: A

Explanation: Each row changes shading from Black -> Grey -> White, keeping the shape consistent. The third row shape is a Triangle, so the missing shape is a White Triangle.

Q9. [MEDIUM] What is the next number in the sequence: 1, 2, 6, 24, 120, ?

- A) 240
- B) 720
- C) 600
- D) 1440

Correct Answer: B

Explanation: This is the factorial sequence: $1! = 1$, $2! = 2$, $3! = 6$, $4! = 24$, $5! = 120$, so the next is $6! = 720$.

Q10. [MEDIUM] Determine the missing number: 8, 12, 24, 60, ?

- A) 120
- B) 180
- C) 150
- D) 240

Correct Answer: B

*Explanation: The multiplier increases by 0.5 at each step: $8 * 1.5 = 12$; $12 * 2.0 = 24$; $24 * 2.5 = 60$; $60 * 3.0 = 180$.*

Q11. [MEDIUM] Find the missing number in the grid: [7 8 56] [9 4 36] [11 ? 66]

- A) 5
- B) 6
- C) 7
- D) 8

Correct Answer: B

*Explanation: In each row, the third number is the product of the first two: $7 * 8 = 56$; $9 * 4 = 36$; $11 * 6 = 66$. So the missing number is 6.*

Q12. [MEDIUM] Complete the letter pattern: AZ, CX, EV, ?

- A) GT
- B) HS
- C) GS
- D) HT

Correct Answer: A

Explanation: First letters increase: $A(+2) \rightarrow C(+2) \rightarrow E(+2) \rightarrow G$. Second letters decrease: $Z(-2) \rightarrow X(-2) \rightarrow V(-2) \rightarrow T$. Thus, GT completes the pattern.

Q13. [MEDIUM] What is the next number: 2, 3, 5, 8, 13, 21, ?

- A) 34
- B) 32
- C) 29
- D) 30

Correct Answer: A

Explanation: This is the Fibonacci sequence, where each number is the sum of the two preceding ones: $13 + 21 = 34$.

Q14. [MEDIUM] In a visual sequence, a clock hand moves by 45 degrees clockwise in step 1, 90 degrees counter-clockwise in step 2, and 135 degrees clockwise in step 3. How much should it move in step 4?

- A) 180 degrees counter-clockwise
- B) 180 degrees clockwise
- C) 220 degrees counter-clockwise
- D) 90 degrees clockwise

Correct Answer: A

Explanation: The angles increase by 45 degrees: 45, 90, 135, so the next is 180 degrees. The direction alternates: CW, CCW, CW, so the next is CCW. Hence, 180 degrees CCW.

Q15. [MEDIUM] Complete the grid pattern: Row 1: [2, 3, 13] (since $2^2 + 3^2 = 13$) Row 2: [4, 1, 17] (since $4^2 + 1^2 = 17$) Row 3: [5, 2, ?]

- A) 27
- B) 29
- C) 31
- D) 35

Correct Answer: B

Explanation: The third column is the sum of squares of the first two columns: $5^2 + 2^2 = 25 + 4 = 29$.

Q16. [ADVANCED] Find the next number in the sequence: 5, 16, 49, 148, ?

- A) 445
- B) 446
- C) 448
- D) 450

Correct Answer: A

*Explanation: The pattern is (previous number * 3) + 1. $(5 * 3) + 1 = 16$. $(16 * 3) + 1 = 49$. $(49 * 3) + 1 = 148$. So $(148 * 3) + 1 = 444 + 1 = 445$.*

Q17. [ADVANCED] Complete the sequence: 0, 4, 18, 48, 100, ?

- A) 180
- B) 150
- C) 200
- D) 175

Correct Answer: A

Explanation: The pattern is $n^3 - n^2$ for $n = 1, 2, 3, 4, 5$. Specifically: $1^3 - 1^2 = 0$; $2^3 - 2^2 = 8 - 4 = 4$; $3^3 - 3^2 = 27 - 9 = 18$; $4^3 - 4^2 = 64 - 16 = 48$; $5^3 - 5^2 = 125 - 25 = 100$; so the next is $6^3 - 6^2 = 216 - 36 = 180$.

Q18. [ADVANCED] In a complex visual matrix, Row 1 has a solid circle with 3 lines. Row 2 has a hollow square with 4 lines. Row 3 has a solid pentagon with 5 lines. What would the next shape logically contain?

- A) Hollow hexagon with 6 lines
- B) Solid hexagon with 6 lines
- C) Hollow hexagon with 5 lines
- D) Solid heptagon with 7 lines

Correct Answer: A

Explanation: The shapes increase in sides: 3 (circle/triangle?), wait - circle with 3 lines, square with 4 lines, pentagon with 5 lines, so the next is a hexagon with 6 lines. The state alternates between solid and hollow: Row 1 solid, Row 2 hollow, Row 3 solid, so Row 4 must be hollow. Thus, a Hollow Hexagon with 6 lines.

Q19. [ADVANCED] What is the missing number in the sequence: 2, 10, 30, 68, 130, ?

- A) 222
- B) 210
- C) 240
- D) 256

Correct Answer: A

Explanation: The pattern is $n^3 + n$ for $n = 1, 2, 3, 4, 5$. Specifically: $1^3+1=2$; $2^3+2=10$; $3^3+3=30$; $4^3+4=68$; $5^3+5=130$; so the next is $6^3+6=216+6=222$.

Q20. [ADVANCED] Determine the next number in this sequence: 6, 24, 60, 120, 210, ?

- A) 336
- B) 343
- C) 320
- D) 360

Correct Answer: A

Explanation: The pattern is $n^3 - n$ for $n = 2, 3, 4, 5, 6$. Specifically: $2^3-2=6$; $3^3-3=24$; $4^3-4=60$; $5^3-5=120$; $6^3-6=210$; so the next is $7^3-7=343-7=336$.

Numerical Intelligence

Q1. [EASY] A shopkeeper sells a product for ₹120, making a profit of 20%. What was the cost price of the product?

- A) ₹100
- B) ₹96
- C) ₹90
- D) ₹105

Correct Answer: A

*Explanation: Selling Price = Cost Price * (1 + Profit%). ₹120 = CP * 1.20. Therefore, CP = 120 / 1.20 = ₹100.*

Q2. [EASY] The average of 5 consecutive numbers is 20. What is the largest of these numbers?

- A) 22
- B) 24
- C) 20
- D) 21

Correct Answer: A

Explanation: Let the numbers be $x-2, x-1, x, x+1, x+2$. Their sum is $5x$, so their average is x . Thus, $x = 20$. The largest number is $x+2 = 22$.

Q3. [EASY] If a car travels at a speed of 60 km/h, how many meters does it cover in one minute?

- A) 1000m
- B) 600m
- C) 1200m
- D) 1500m

Correct Answer: A

Explanation: Speed is 60 km/h = 60,000 meters / 60 minutes = 1000 meters/minute. So in one minute, it covers 1000 meters.

Q4. [EASY] Solve for X: $3X + 7 = 28 - 4X$.

- A) 3
- B) 4
- C) 5
- D) 7

Correct Answer: A

Explanation: Combine X terms: $3X + 4X = 28 - 7 \Rightarrow 7X = 21 \Rightarrow X = 3$.

Q5. [EASY] What is 15% of 200 plus 25% of 80?

- A) 50
- B) 45
- C) 60
- D) 40

Correct Answer: A

Explanation: 15% of 200 = 30. 25% of 80 = 20. $30 + 20 = 50$.

Q6. [EASY] If a triangle has a base of 12 cm and a height of 8 cm, what is its area?

- A) 48 sq cm
- B) 96 sq cm
- C) 24 sq cm
- D) 40 sq cm

Correct Answer: A

*Explanation: Area of a triangle = $0.5 * \text{base} * \text{height} = 0.5 * 12 * 8 = 48 \text{ sq cm}$.*

Q7. [MEDIUM] A sum of money doubles itself in 8 years at simple interest. In how many years will it triple itself at the same rate?

- A) 16 years
- B) 12 years
- C) 24 years
- D) 20 years

Correct Answer: A

*Explanation: For a sum P to double (become 2P), the simple interest earned must be P. This takes 8 years. For the sum to triple (become 3P), the simple interest earned must be 2P. At the same rate, earning double interest will take double time: $8 * 2 = 16$ years.*

Q8. [MEDIUM] A train 150 meters long passes a telegraph post in 12 seconds. How long will it take to cross a bridge of length 250 meters?

- A) 20 seconds
- B) 32 seconds
- C) 24 seconds
- D) 30 seconds

Correct Answer: B

Explanation: Speed of the train = distance / time = $150\text{m} / 12\text{s} = 12.5 \text{ m/s}$. To cross the bridge, the total distance to cover is train length + bridge length = $150 + 250 = 400$ meters. Time taken = $400 / 12.5 = 32$ seconds.

Q9. [MEDIUM] A can do a piece of work in 10 days, and B can do it in 15 days. They work together for 4 days, then A leaves. How many more days will B take to finish the remaining work?

- A) 5 days
- B) 6 days
- C) 4 days
- D) 3 days

Correct Answer: A

*Explanation: Work rate of A = $1/10$, B = $1/15$. Combined rate = $1/10 + 1/15 = 5/30 = 1/6$. In 4 days they complete $4 * (1/6) = 2/3$ of the work. Remaining work = $1 - 2/3 = 1/3$. B finishes this remaining work in $(1/3) / (1/15) = 5$ days.*

Q10. [MEDIUM] The ratio of ages of a father and his son is 7:3. If the sum of their ages is 60 years, what will be the ratio of their ages after 6 years?

- A) 2:1
- B) 3:1
- C) 8:3
- D) 5:2

Correct Answer: A

Explanation: Father's age = $7x$, Son's age = $3x$. Sum is $10x = 60 \Rightarrow x = 6$. Father's age is 42, son's age is 18. After 6 years, father is 48 and son is 24. Ratio = $48:24 = 2:1$.

Q11. [MEDIUM] In a class of 60 students, 40% are girls. How many boys are there in the class?

- A) 36
- B) 24
- C) 30
- D) 40

Correct Answer: A

Explanation: Girls = 40% of 60 = 24. Boys = $60 - 24 = 36$.

Q12. [MEDIUM] What is the compound interest on $\text{₹}10,000$ for 2 years at 10% per annum, compounded annually?

- A) $\text{₹}2,100$
- B) $\text{₹}2,000$
- C) $\text{₹}1,900$
- D) $\text{₹}2,200$

Correct Answer: A

*Explanation: Amount = $10,000 * (1 + 0.10)^2 = 10,000 * 1.21 = \text{₹}12,100$. Interest = $12,100 - 10,000 = \text{₹}2,100$.*

Q13. [MEDIUM] A container contains 40 liters of milk. From this container, 4 liters of milk were taken out and replaced by water. This process was repeated further two times. How much milk is now contained by the container?

- A) 29.16 liters
- B) 32.4 liters
- C) 28.5 liters
- D) 30 liters

Correct Answer: A

*Explanation: Milk left = Original quantity * $(1 - \text{taken_out/original})^n = 40 * (1 - 4/40)^3 = 40 * (0.9)^3 = 40 * 0.729 = 29.16$ liters.*

Q14. [MEDIUM] Find the value of X if $\log(X) + \log(5) = 2$.

- A) 20
- B) 10
- C) 50
- D) 100

Correct Answer: A

Explanation: $\log(X) + \log(5) = \log(5X) = 2$. Assuming base 10: $5X = 10^2 = 100$. Therefore, $X = 20$.

Q15. [ADVANCED] A boat goes 24 km upstream and 28 km downstream in 6 hours. It goes 30 km upstream and 21 km downstream in 6.5 hours. What is the speed of the boat in still water?

- A) 10 km/h
- B) 8 km/h
- C) 12 km/h
- D) 14 km/h

Correct Answer: A

Explanation: Let upstream speed be u and downstream speed be d . Equations: $24/u + 28/d = 6$, and $30/u + 21/d = 6.5$. Solving these: let $1/u = x$ and $1/d = y$. $24x + 28y = 6$ and $30x + 21y = 6.5$. Multiply first by 3 and second by 4: $72x + 84y = 18$ and $120x + 84y = 26$. Subtracting: $48x = 8 \Rightarrow x = 1/6 \Rightarrow u = 6$ km/h. Substitute x in first: $4 + 28y = 6 \Rightarrow 28y = 2 \Rightarrow y = 1/14 \Rightarrow d = 14$ km/h. Speed in still water = $(d + u)/2 = (14 + 6)/2 = 10$ km/h.

Q16. [ADVANCED] A bag contains 5 red, 4 green, and 3 blue balls. If three balls are drawn at random, what is the probability that they are of different colors?

- A) $3/11$
- B) $5/11$
- C) $6/11$
- D) $4/11$

Correct Answer: A

Explanation: Total balls = 12. Number of ways to draw 3 balls = ${}^{12}C_3 = 220$. Ways to draw 1 ball of each color = ${}^5C_1 * {}^4C_1 * {}^3C_1 = 5 * 4 * 3 = 60$. Probability = $60 / 220 = 3/11$.

Q17. [ADVANCED] The product of two positive integers is 2024. If their greatest common divisor (GCD) is 2, what is their least common multiple (LCM)?

- A) 1012
- B) 2024
- C) 506
- D) 4048

Correct Answer: A

Explanation: For any two numbers A and B : $A * B = \text{GCD}(A, B) * \text{LCM}(A, B)$. $2024 = 2 * \text{LCM}$. $\text{LCM} = 2024 / 2 = 1012$.

Q18. [ADVANCED] How many integer solutions exist for the inequality: $x^2 - 8x + 12 < 0$?

- A) 3
- B) 5
- C) 2
- D) 4

Correct Answer: A

Explanation: Factor the quadratic: $(x - 2)(x - 6) < 0$. This inequality holds true for $2 < x < 6$. The integers in this range are 3, 4, and 5. There are 3 integer solutions.

Q19. [ADVANCED] If $\log_2(x) + \log_4(x) + \log_{16}(x) = 7$, find the value of x .

- A) 16
- B) 64
- C) 32
- D) 8

Correct Answer: A

Explanation: Change base to 2: $\log_2(x) + 0.5 \cdot \log_2(x) + 0.25 \cdot \log_2(x) = 7 \Rightarrow (1 + 0.5 + 0.25) \cdot \log_2(x) = 7 \Rightarrow 1.75 \cdot \log_2(x) = 7 \Rightarrow (7/4) \cdot \log_2(x) = 7 \Rightarrow \log_2(x) = 4 \Rightarrow x = 2^4 = 16$.

Q20. [ADVANCED] Find the sum of the infinite geometric series: $9 - 3 + 1 - 1/3 + \dots$

- A) 6.75
- B) 6.25
- C) 7.5
- D) 9

Correct Answer: A

Explanation: This is a geometric series with first term $a = 9$ and common ratio $r = -1/3$. The sum $S = a / (1 - r) = 9 / (1 - (-1/3)) = 9 / (4/3) = 27/4 = 6.75$.

Verbal Reasoning

Q1. [EASY] Choose the word that is most nearly opposite in meaning to: OBSTINATE.

- A) Flexible
- B) Stubborn
- C) Rigid
- D) Callous

Correct Answer: A

Explanation: Obstinate means stubborn or refusing to change one's opinion. The opposite is flexible.

Q2. [EASY] Find the analogy: BREAD : FLOUR :: PAPER : ?

- A) Wood
- B) Book
- C) Pen
- D) Tree

Correct Answer: A

Explanation: Flour is the primary raw material used to make bread. Similarly, wood (pulp) is the raw material used to make paper.

Q3. [EASY] Which of the following spelling is correct?

- A) Accommodate
- B) Acomodate
- C) Accomodate
- D) Acommadate

Correct Answer: A

Explanation: The correct spelling is "Accommodate" with double c and double m.

Q4. [EASY] Complete the sentence: The manager decided to _____ the meeting until next Monday due to scheduling conflicts.

- A) postpone
- B) cancel
- C) prepone
- D) convene

Correct Answer: A

Explanation: To delay a meeting to a later date is to postpone it. Prepone means to make it earlier, which is incorrect in this context.

Q5. [EASY] Identify the synonym of: MITIGATE.

- A) Alleviate
- B) Aggravate
- C) Incite
- D) Prolong

Correct Answer: A

Explanation: Mitigate means to make less severe, serious, or painful. Alleviate is a synonym.

Q6. [EASY] Identify the odd word: Listening, Hearing, Speaking, Attending.

- A) Speaking
- B) Listening
- C) Hearing
- D) Attending

Correct Answer: A

Explanation: Listening, hearing, and attending are receptive actions related to receiving communication, while speaking is expressive.

Q7. [EASY] Choose the option that describes the idiom: "Spill the beans".

- A) To reveal a secret
- B) To make a mess
- C) To waste food
- D) To cook a meal

Correct Answer: A

Explanation: The idiom "spill the beans" means to disclose secret information prematurely or unintentionally.

Q8. [MEDIUM] Choose the word that best fits the blank: Despite the speaker's _____ tone, the audience remained skeptical of his proposed reforms.

- A) persuasive
- B) hostile
- C) monotonous
- D) ambiguous

Correct Answer: A

Explanation: The word "Despite" signals a contrast. Since the audience was skeptical, the speaker must have had an appealing or convincing tone, which fits "persuasive".

Q9. [MEDIUM] What is the correct order of the sentences to form a coherent paragraph? P: In the long run, this leads to environmental degradation. Q: Industrialization has brought about significant economic growth. R: However, it has also caused massive pollution. S: Clean energy solutions must be adopted to balance progress and ecology.

- A) QRPS
- B) QPRS
- C) PRQS
- D) QSPR

Correct Answer: A

Explanation: Q starts with the positive impact (growth). R introduces a contrast ("However... pollution"). P extends the consequence of pollution ("In the long run... degradation"). S provides the solution ("Clean energy..."). So QRPS is the most coherent sequence.

Q10. [MEDIUM] Choose the option that is most similar in meaning to the capitalized word: EPHEMERAL.

- A) Short-lived
- B) Eternal
- C) Mythical
- D) Vibrant

Correct Answer: A

Explanation: Ephemeral means lasting for a very short time. Short-lived is the closest synonym.

Q11. [MEDIUM] Choose the word that is OPPOSITE to: PRAGMATIC.

- A) Idealistic
- B) Practical
- C) Sensible
- D) Realistic

Correct Answer: A

Explanation: Pragmatic means dealing with things sensibly and realistically in a way that is based on practical considerations. The opposite is idealistic.

Q12. [MEDIUM] Choose the sentence that is grammatically correct.

- A) Neither of the candidates has submitted their reports.
- B) Neither of the candidates have submitted their reports.
- C) Neither of the candidate has submitted their reports.
- D) Neither of the candidates has submitted his or her report.

Correct Answer: D

Explanation: "Neither" is singular, so it requires a singular verb ("has"). It also requires singular pronouns ("his or her" instead of "their"). Thus, option D is the most correct.

Q13. [MEDIUM] Identify the analogy: GARRULOUS : TALKATIVE :: TACITURN : ?

- A) Reserved
- B) Loud
- C) Eloquent
- D) Verbose

Correct Answer: A

Explanation: Garrulous is a synonym for talkative. Taciturn refers to a person who is reserved or says very little. Thus, reserved is the correct synonym.

Q14. [MEDIUM] Complete the sentence: Her _____ memory allowed her to recall complex legal briefs down to the exact page number.

- A) eidetic
- B) evasive
- C) erratic
- D) elusive

Correct Answer: A

Explanation: Eidetic memory (photographic memory) refers to the ability to recall images, sounds, or objects in memory with extreme precision. Other words do not fit.

Q15. [MEDIUM] What is the meaning of the idiom "To play devil's advocate"?

- A) To argue for an unpopular opinion for the sake of debate
- B) To support a bad person
- C) To behave mischievously
- D) To give legal advice to a criminal

Correct Answer: A

Explanation: Playing devil's advocate means expressing a viewpoint that disagrees with the consensus to provoke further discussion or test the strength of the argument.

Q16. [ADVANCED] Select the word that best fits both blanks: (1) The new policy is designed to _____ the conflict. (2) Scientists are trying to _____ the effects of the chemical spill.

- A) mitigate
- B) exacerbate
- C) escalate
- D) evaluate

Correct Answer: A

Explanation: "Mitigate" means to make less severe. It fits both sentences: mitigating (reducing) a conflict, and mitigating (minimizing) the damage from a chemical spill.

Q17. [ADVANCED] Read the statement: "Most successful entrepreneurs have failed at least once before making it big." Which of the following is an assumption implicit in the statement?

- A) Failure is a necessary prerequisite for entrepreneurship success.
- B) Some entrepreneurs succeed without experiencing initial failure.
- C) Entrepreneurs who fail never try again.
- D) Successful entrepreneurs have higher risk tolerances than regular managers.

Correct Answer: B

Explanation: The statement says "Most" successful entrepreneurs have failed, which logically implies that some (the minority) did not fail first. Thus, option B is an implicit truth. A is too extreme ("necessary prerequisite").

Q18. [ADVANCED] Choose the word that is closest to the meaning of: ANACHRONISM.

- A) A thing belonging to a period other than that in which it exists
- B) A severe disease causing memory loss
- C) A grammatical error in academic writing
- D) A mathematical formula used in mapping timelines

Correct Answer: A

Explanation: An anachronism is a chronological inconsistency, especially placing something in a time period where it does not belong.

Q19. [ADVANCED] Identify the sentence with a misplaced modifier.

- A) Walking down the street, the library appeared on my left.
- B) While walking down the street, I saw the library on my left.
- C) The library appeared on my left as I walked down the street.
- D) As I was walking down the street, the library appeared on my left.

Correct Answer: A

Explanation: In A, "Walking down the street" is a dangling modifier because it seems to describe "the library" (which cannot walk). In the other options, the modifier clearly attaches to "I".

Q20. [ADVANCED] Find the pair of words that completes the analogy: CACOPHONY : EUPHONY :: _____ :

- A) Noxious : Wholesome
- B) Quiet : Silent
- C) Harsh : Unpleasant
- D) Beautiful : Pretty

Correct Answer: A

Explanation: Cacophony (harsh noise) and Euphony (sweet sound) are antonyms representing bad vs good quality. Noxious (harmful) and Wholesome (healthy) are similarly antonyms representing bad vs good.

Analytical Thinking

Q1. [EASY] A family has two parents and three children. In how many ways can they stand in a line for a photograph if the parents must stand on the two ends?

- A) 12
- B) 6
- C) 24
- D) 120

Correct Answer: A

*Explanation: The parents can stand at the two ends in $2! = 2$ ways. The 3 children can stand in the middle 3 seats in $3! = 6$ ways. Total arrangements = $2 * 6 = 12$.*

Q2. [EASY] In a group of 30 people, 18 drink tea and 15 drink coffee. If everyone drinks at least one of the two, how many drink both tea and coffee?

- A) 3
- B) 5
- C) 8
- D) 12

Correct Answer: A

Explanation: Total people = Tea + Coffee - Both. $30 = 18 + 15 - \text{Both} \Rightarrow \text{Both} = 33 - 30 = 3$.

Q3. [EASY] If a box contains 3 red, 4 white, and 5 blue chips, what is the minimum number of chips you must pick to guarantee that you have at least two chips of the same color?

- A) 4
- B) 3
- C) 5
- D) 6

Correct Answer: A

Explanation: By the Pigeonhole Principle, since there are 3 colors, picking 4 chips guarantees at least two will share a color.

Q4. [EASY] If X is greater than Y, and Y is equal to Z, which statement is definitely false?

- A) X is less than Z
- B) X is greater than Z
- C) Y is less than X
- D) Z is less than X

Correct Answer: A

Explanation: Since $Y = Z$, substituting Z for Y in " $X > Y$ " gives $X > Z$. Therefore, "X is less than Z" is definitely false.

Q5. [EASY] A list contains numbers [3, 7, 15, 31, X]. What is the value of X?

- A) 63
- B) 47
- C) 59
- D) 61

Correct Answer: A

*Explanation: The pattern is $2n + 1$: $3*2+1=7$; $7*2+1=15$; $15*2+1=31$; so $31*2+1=63$.*

Q6. [EASY] In a Venn diagram of two overlapping sets A and B, Set A has 10 elements, Set B has 15 elements, and they share 5 elements. How many elements are in Set A but NOT Set B?

- A) 5
- B) 10
- C) 15
- D) 20

Correct Answer: A

Explanation: Elements in A only = Total in A - Shared elements = $10 - 5 = 5$.

Q7. [MEDIUM] If 4 workers can pack 4 boxes in 4 minutes, how many workers are needed to pack 20 boxes in 20 minutes?

- A) 4
- B) 20
- C) 10
- D) 5

Correct Answer: A

*Explanation: Using the formula $(\text{Workers} * \text{Time}) / \text{Work} = \text{Constant}$. $(4 * 4) / 4 = (W * 20) / 20$. This simplifies to $4 = W$. So, 4 workers are needed.*

Q8. [MEDIUM] A company has three departments: Sales, IT, and HR. Out of 100 employees, 50 are in Sales, 40 are in IT, and 20 are in HR. If 10 employees work in both Sales and IT, and none work in HR and any other department, how many employees are there in total? (Wait, does the total match?)

- A) 100
- B) 90
- C) 110
- D) 95

Correct Answer: A

Explanation: Sales only = $50 - 10 = 40$. IT only = $40 - 10 = 30$. Sales & IT = 10. HR = 20. Total = $40 + 30 + 10 + 20 = 100$.

Q9. [MEDIUM] A dice is rolled twice. What is the probability that the sum of the two rolls is at least 10?

- A) $1/6$
- B) $1/12$
- C) $5/36$
- D) $1/4$

Correct Answer: A

Explanation: Total outcomes = 36. Outcomes with sum ≥ 10 are (4,6), (5,5), (5,6), (6,4), (6,5), (6,6). There are 6 such outcomes. Probability = $6/36 = 1/6$.

Q10. [MEDIUM] In a code, "APPLE" is written as "5-16-16-12-5" based on letter positions. How is "ORANGE" written?

- A) 15-18-1-14-7-5
- B) 15-17-1-14-8-5
- C) 14-18-1-13-7-5
- D) 15-18-2-14-7-5

Correct Answer: A

Explanation: The code simply replaces each letter with its index in the alphabet: O=15, R=18, A=1, N=14, G=7, E=5. Hence, 15-18-1-14-7-5.

Q11. [MEDIUM] If you have a 3-liter jug and a 5-liter jug, and an unlimited supply of water, what is the minimum number of steps to measure exactly 4 liters? (A step is filling, pouring, or emptying a jug).

- A) 6
- B) 5
- C) 7
- D) 8

Correct Answer: A

Explanation: Steps: (1) Fill 5L jug. (2) Pour from 5L to fill 3L (leaves 2L in 5L). (3) Empty 3L jug. (4) Pour 2L from 5L to 3L jug. (5) Fill 5L jug. (6) Pour from 5L to fill remaining space in 3L jug (needs 1L, leaving exactly 4L in 5L jug). Total steps = 6.

Q12. [MEDIUM] A group of students took a test. 80% passed in Math, 75% passed in English, and 70% passed in both. What percent of students failed in both?

- A) 15%
- B) 10%
- C) 20%
- D) 5%

Correct Answer: A

Explanation: Percent passing at least one = Math + English - Both = $80\% + 75\% - 70\% = 85\%$. Percent failing both

$$= 100\% - 85\% = 15\%.$$

Q13. [MEDIUM] A data table shows product sales: Q1: \$10k, Q2: \$12k, Q3: \$15k, Q4: \$18k. What is the percentage increase in sales from Q1 to Q4?

- A) 80%
- B) 40%
- C) 60%
- D) 50%

Correct Answer: A

*Explanation: Increase = $Q4 - Q1 = \$18k - \$10k = \$8k$. Percentage increase = $(8k / 10k) * 100 = 80\%$.*

Q14. [MEDIUM] If $A > B$, $B < C$, and $C = D$, which of the following is always true?

- A) $B < D$
- B) $A > C$
- C) $A < D$
- D) $B > D$

Correct Answer: A

Explanation: Since $B < C$ and $C = D$, substituting D for C gives $B < D$. This is always true.

Q15. [MEDIUM] How many rectangles are there in a standard 3x3 grid of squares?

- A) 36
- B) 9
- C) 14
- D) 30

Correct Answer: A

*Explanation: A rectangle is defined by choosing 2 horizontal lines and 2 vertical lines. In a 3x3 grid of squares, there are 4 horizontal lines and 4 vertical lines. Number of rectangles = $4C2 * 4C2 = 6 * 6 = 36$.*

Q16. [ADVANCED] A software system fails if any of its 3 independent components fails. If the probability of failure for components A, B, and C is 0.1, 0.2, and 0.05 respectively, what is the probability that the system functions successfully?

- A) 0.684
- B) 0.650
- C) 0.720
- D) 0.900

Correct Answer: A

Explanation: The probability of functioning successfully is the product of individual success probabilities:

$$P(\text{success}) = P(A \text{ works}) * P(B \text{ works}) * P(C \text{ works}) = (1 - 0.1) * (1 - 0.2) * (1 - 0.05) = 0.9 * 0.8 * 0.95 = 0.72 * 0.95 = 0.684.$$

Q17. [ADVANCED] In a survey of 100 students, 60 like Math, 50 like Science, and 30 like History. 20 like both Math and Science, 15 like Science and History, 10 like Math and History. 5 like all three. How many students like NONE of the three subjects?

- A) 0
- B) 5
- C) 10
- D) 15

Correct Answer: A

Explanation: Using the union formula for 3 sets: $N(M \cup S \cup H) = N(M) + N(S) + N(H) - N(M \cap S) - N(S \cap H) - N(M \cap H) + N(M \cap S \cap H) = 60 + 50 + 30 - 20 - 15 - 10 + 5 = 140 - 45 + 5 = 100$. Since all 100 students like at least one, 0 students like none.

Q18. [ADVANCED] If $\log(y) = 3 * \log(x) + \log(2)$, which equation expresses y in terms of x?

- A) $y = 2x^3$
- B) $y = x^3 + 2$
- C) $y = 8x^3$
- D) $y = 6x$

Correct Answer: A

*Explanation: $\log(y) = \log(x^3) + \log(2) \Rightarrow \log(y) = \log(2 * x^3)$. Removing logs gives $y = 2x^3$.*

Q19. [ADVANCED] How many diagonals does a regular decagon (10-sided polygon) have?

- A) 35
- B) 45
- C) 20
- D) 50

Correct Answer: A

*Explanation: Number of diagonals = $n(n - 3) / 2 = 10 * 7 / 2 = 35$.*

Q20. [ADVANCED] In a binary search tree, what is the maximum number of comparisons needed to find a node if the tree is balanced and contains 31 nodes?

- A) 5
- B) 31
- C) 6
- D) 4

Correct Answer: A

Explanation: For a balanced BST with N nodes, the maximum comparisons is $\log_2(N+1)$. $\log_2(32) = 5$.

Problem Solving

Q1. [EASY] If a water tank has two pipes, one can fill it in 3 hours and the other can empty it in 6 hours. If both pipes are opened together, how long will it take to fill the empty tank?

- A) 6 hours
- B) 4 hours
- C) 5 hours
- D) 8 hours

Correct Answer: A

Explanation: Net rate = Fill rate - Empty rate = $1/3 - 1/6 = 1/6$. Thus, it will take 6 hours to fill the tank.

Q2. [EASY] If you double the radius of a circle, by what factor does its area increase?

- A) 4
- B) 2
- C) 8
- D) 16

Correct Answer: A

*Explanation: Area of circle = $\pi * r^2$. If r becomes $2r$, new Area = $\pi * (2r)^2 = 4 * \pi * r^2$. The area increases by a factor of 4.*

Q3. [EASY] A farmer has chickens and rabbits. If there are 35 heads and 94 legs in total, how many rabbits does the farmer have?

- A) 12
- B) 23
- C) 15
- D) 20

Correct Answer: A

Explanation: Let C be chickens and R be rabbits. $C + R = 35$, and $2C + 4R = 94$. Multiply the first equation by 2: $2C + 2R = 70$. Subtracting this from the second: $2R = 24 \Rightarrow R = 12$.

Q4. [EASY] If a smartphone costs ₺15,000 after a 10% discount, what was its original price?

- A) ₺16,667
- B) ₺16,500
- C) ₺16,000
- D) ₺17,000

Correct Answer: A

*Explanation: Selling Price = Original Price * 0.90. $₺15,000 = \text{Original Price} * 0.90 \Rightarrow \text{Original Price} = 15,000 / 0.90 = ₺16,666.67$.*

Q5. [EASY] If you roll a standard 6-sided die, what is the probability of rolling an even number?

- A) $1/2$
- B) $1/3$
- C) $2/3$
- D) $1/6$

Correct Answer: A

Explanation: Even numbers are 2, 4, and 6. That is 3 out of 6 options, so the probability is $3/6 = 1/2$.

Q6. [EASY] If a computer program executes 100 operations in 0.05 seconds, how many operations can it perform in 2 seconds at the same rate?

- A) 4000
- B) 2000
- C) 1000
- D) 5000

Correct Answer: A

*Explanation: Rate = $100 / 0.05 = 2000$ operations per second. In 2 seconds: $2000 * 2 = 4000$ operations.*

Q7. [MEDIUM] A cylindrical container of radius 7 cm and height 10 cm is filled with water. If we pour this water into a rectangular block container of base dimensions 11 cm x 14 cm, what will be the height of water in the rectangular container?

- A) 10 cm
- B) 5 cm
- C) 7 cm
- D) 8 cm

Correct Answer: A

*Explanation: Volume of cylinder = $\pi * r^2 * h = (22/7) * 7 * 7 * 10 = 1540$ cubic cm. Volume of rectangular block = $base_area * height = 11 * 14 * h_{rect} = 154 * h_{rect}$. Equating volumes: $1540 = 154 * h_{rect} \Rightarrow h_{rect} = 10$ cm.*

Q8. [MEDIUM] Three light bulbs flash at intervals of 4 seconds, 6 seconds, and 10 seconds. If they all flash together now, after how many seconds will they next flash together?

- A) 60 seconds
- B) 120 seconds
- C) 40 seconds
- D) 80 seconds

Correct Answer: A

Explanation: They will flash together at intervals equal to the Least Common Multiple (LCM) of 4, 6, and 10. $LCM(4, 6) = 12$. $LCM(12, 10) = 60$. So, they flash together every 60 seconds.

Q9. [MEDIUM] If a principal of ₹800 amounts to ₹920 in 3 years at simple interest, what will it amount to if the rate of interest is increased by 3%?

- A) ₹992
- B) ₹956
- C) ₹972
- D) ₹1,000

Correct Answer: A

*Explanation: Interest earned = $₹920 - ₹800 = ₹120$ in 3 years. Interest per year = ₹40. Rate = $(40 / 800) * 100 = 5\%$. New rate = $5\% + 3\% = 8\%$. New interest in 3 years = $800 * 8\% * 3 = ₹192$. New amount = $800 + 192 = ₹992$.*

Q10. [MEDIUM] A student has to secure 40% marks to pass. He gets 178 marks and fails by 22 marks. What are the maximum marks of the exam?

- A) 500
- B) 400
- C) 450
- D) 600

Correct Answer: A

Explanation: Pass marks = Marks scored + failing margin = $178 + 22 = 200$ marks. 40% of Maximum Marks = 200 $\Rightarrow$ Maximum Marks = $200 / 0.40 = 500$.

Q11. [MEDIUM] In what ratio must water be mixed with milk costing ₹12 per liter to obtain a mixture worth ₹8 per liter?

- A) 1:2
- B) 1:3
- C) 2:3
- D) 1:1

Correct Answer: A

Explanation: Let cost of water be ₹0. Milk is ₹12. Target mixture price is ₹8. By rule of alligation: (Water : Milk) = $(12 -$

$$8) : (8 - 0) = 4 : 8 = 1 : 2.$$

Q12. [MEDIUM] The difference between the squares of two consecutive odd integers is always divisible by:

- A) 8
- B) 3
- C) 6
- D) 7

Correct Answer: A

Explanation: Let the consecutive odd integers be $2n+1$ and $2n-1$. Difference of squares = $(2n+1)^2 - (2n-1)^2 = (4n^2 + 4n + 1) - (4n^2 - 4n + 1) = 8n$. This is always divisible by 8.

Q13. [MEDIUM] How many ways are there to choose a committee of 3 people from a group of 6 people?

- A) 20
- B) 18
- C) 15
- D) 12

Correct Answer: A

*Explanation: Number of combinations = $6C3 = (6 * 5 * 4) / (3 * 2 * 1) = 20$.*

Q14. [MEDIUM] A clock gains 5 minutes every hour. If it is set correctly at 8:00 AM, what time will it show when the actual time is 2:00 PM on the same day?

- A) 2:30 PM
- B) 2:25 PM
- C) 2:35 PM
- D) 2:20 PM

Correct Answer: A

*Explanation: From 8:00 AM to 2:00 PM is 6 hours. Since it gains 5 minutes every hour, in 6 hours it will gain $6 * 5 = 30$ minutes. The clock will show 2:30 PM.*

Q15. [MEDIUM] If 3 machines can manufacture 150 widgets in 5 hours, how many widgets can 5 machines manufacture in 6 hours?

- A) 300
- B) 250
- C) 280
- D) 320

Correct Answer: A

*Explanation: Widget rate per machine per hour = $150 / (3 * 5) = 150 / 15 = 10$ widgets. In 6 hours, 5 machines will manufacture = $5 * 6 * 10 = 300$ widgets.*

Q16. [ADVANCED] A sphere is inscribed inside a cylinder of radius R and height $2R$. What is the ratio of the volume of the sphere to the volume of the cylinder?

- A) 2:3
- B) 1:2
- C) 3:4
- D) 4:5

Correct Answer: A

*Explanation: Volume of sphere = $(4/3) * \pi * R^3$. Volume of cylinder = $\pi * R^2 * h = \pi * R^2 * (2R) = 2 * \pi * R^3$.*

$$\text{Ratio} = ((4/3) * \pi * R^3) / (2 * \pi * R^3) = (4/3) / 2 = 2/3.$$

Q17. [ADVANCED] If $x + y + z = 12$ and $xy + yz + zx = 47$, find the value of $x^2 + y^2 + z^2$.

- A) 50
- B) 47
- C) 44
- D) 54

Correct Answer: A

Explanation: Using the algebraic identity: $(x+y+z)^2 = x^2+y^2+z^2 + 2(xy+yz+zx)$. $12^2 = x^2+y^2+z^2 + 2(47) \Rightarrow 144 = x^2+y^2+z^2 + 94 \Rightarrow x^2+y^2+z^2 = 144 - 94 = 50$.

Q18. [ADVANCED] A path of uniform width 2m surrounds a rectangular lawn of dimensions 20m x 15m. What is the area of the path?

- A) 156 sq m
- B) 140 sq m
- C) 160 sq m
- D) 120 sq m

Correct Answer: A

*Explanation: Area of lawn = $20 * 15 = 300$ sq m. The outer dimensions including the path are $(20 + 2*2)$ by $(15 + 2*2) = 24$ by 19 m. Outer area = $24 * 19 = 456$ sq m. Area of path = Outer area - lawn area = $456 - 300 = 156$ sq m.*

Q19. [ADVANCED] How many positive integers less than 1000 are divisible by neither 5 nor 7?

- A) 686
- B) 685
- C) 680
- D) 700

Correct Answer: A

Explanation: Integers less than 1000 (from 1 to 999). Divisible by 5: $\text{floor}(999/5) = 199$. Divisible by 7: $\text{floor}(999/7) = 142$. Divisible by both 35: $\text{floor}(999/35) = 28$. Divisible by 5 or 7 = $199 + 142 - 28 = 313$. Divisible by neither = $999 - 313 = 686$.

Q20. [ADVANCED] Find the units digit of the number 3^{2026} .

- A) 9
- B) 3
- C) 7
- D) 1

Correct Answer: A

Explanation: The units digit of powers of 3 repeats in a cycle of 4: $3^1=3$, $3^2=9$, $3^3=7$, $3^4=1$. Divide the exponent 2026 by 4: remainder is 2. The units digit corresponds to 3^2 , which is 9.
